

(ii) Calculate the centripetal acceleration of star A.

acceleration = ms^{-2} [1]

(iii) Use your answer in (b)(ii) to determine the period of the orbit of star A.

period = s [3]

(iv) By placing a tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for star B compare with the same quantities for star A.

Table 1.1

	B less than A	B equal to A	B greater than A
centripetal acceleration			
linear speed			
period			

[3]

[Total: 11]

